Experiment No :3

Tutorial Title: Perform Bubble Sort, Selection Sort, Insertion Sort, Quick Sort, and Merge Sort using C

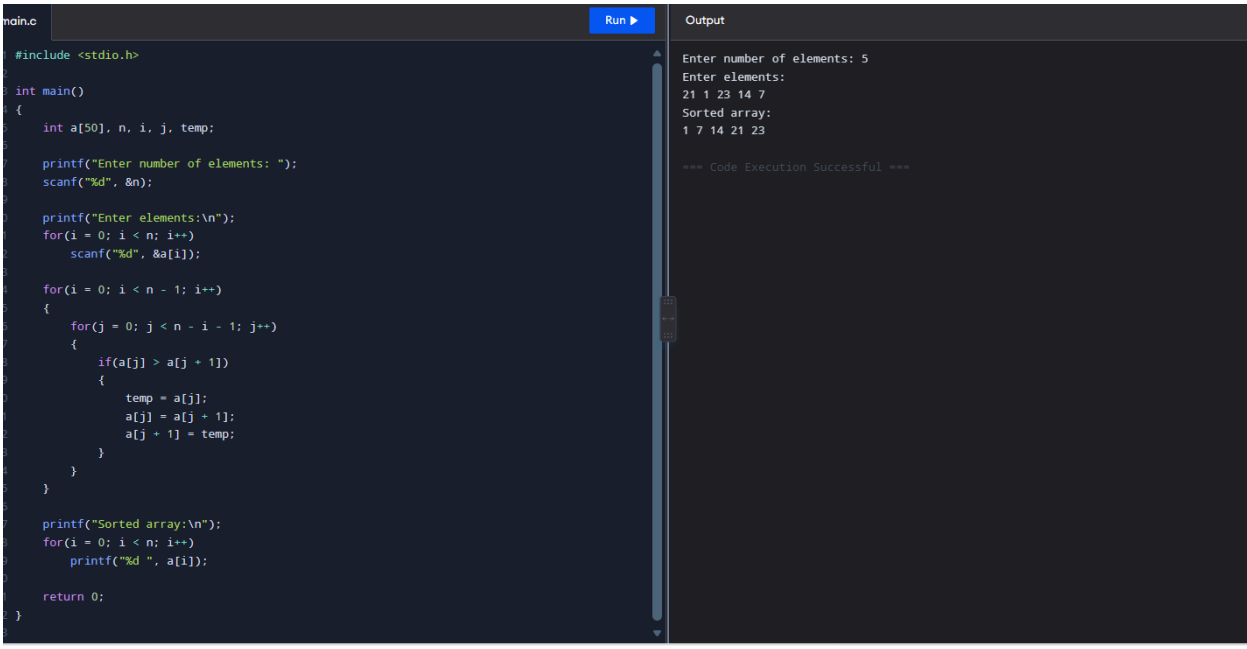
Name of Student: Lohakare Shweta

Class: SY **Div:** 2 **PRN No:**B25ET1099 **Batch:**B

Program: Bubble sort

```
#include <stdio.h>
int main()
{
   int a[50], n, i, j, temp;
  printf("Enter number of elements: ");
   scanf("%d", &n);
  printf("Enter elements:\n");
   for(i = 0; i < n; i++)
      scanf("%d", &a[i]);
  for(i = 0; i < n - 1; i++)
   {
      for(j = 0; j < n - i - 1; j++)
      {
         if(a[j] > a[j + 1])
         {
            temp = a[j];
            a[j] = a[j + 1];
            a[j + 1] = temp;
         }
      }
   }
  printf("Sorted array:\n");
   for(i = 0; i < n; i++)
      printf("%d ", a[i]);
  return 0; }
```

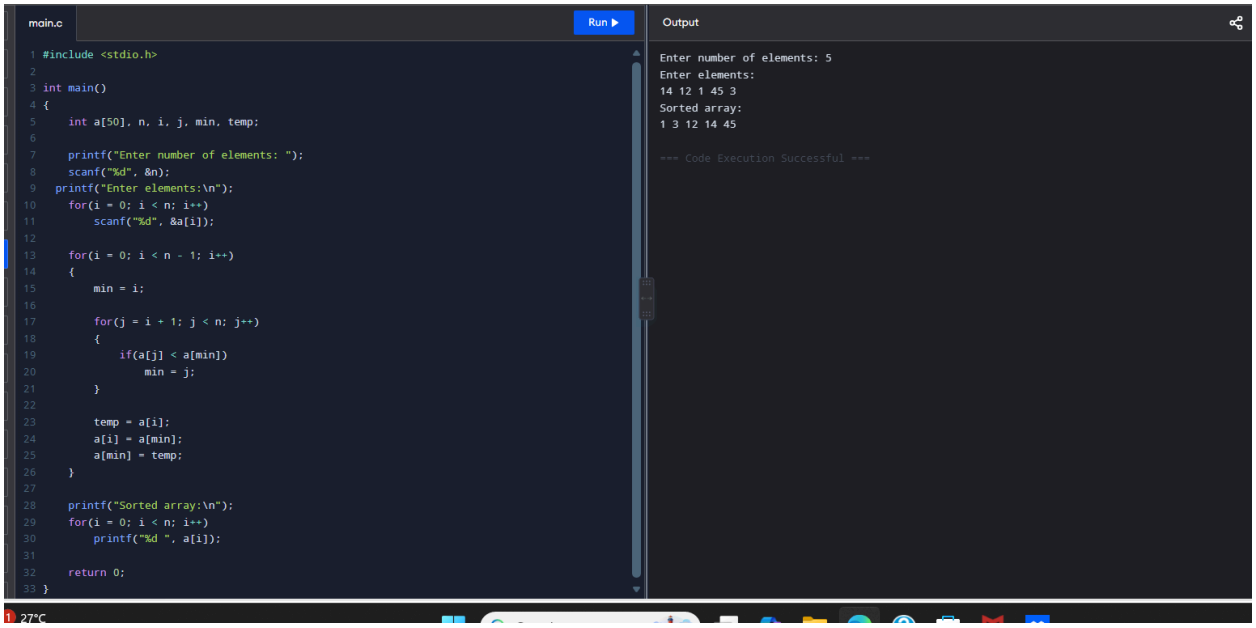
Output:



Program: Selection sort

```
#include <stdio.h>
int main()
{
   int a[50], n, i, j, min, temp;
  printf("Enter number of elements: ");
   scanf("%d", &n);
  printf("Enter elements:\n");
   for(i = 0; i < n; i++)
      scanf("%d", &a[i]);
  for(i = 0; i < n - 1; i++)
   {
      min = i;
      for(j = i + 1; j < n; j++)
      {
         if(a[j] < a[min])
            min = j; }
        temp = a[i];
      a[i] = a[min];
      a[min] = temp; }
     printf("Sorted array:\n");
   for(i = 0; i < n; i++)
      printf("%d ", a[i]);
  return 0;  }
```

Output:



Program: Insertion sort

```
#include <stdio.h>
 int main()
{
   int a[50], n, i, j, key;
   printf("Enter number of elements: ");
   scanf("%d", &n);
  printf("Enter elements:\n");
   for(i = 0; i < n; i++)
      scanf("%d", &a[i]);
   for(i = 1; i < n; i++)
   {   key = a[i];
      j = i - 1;
while(j >= 0 && a[j] > key)
      {   a[j + 1] = a[j];
         j--;
      }
a[j + 1] = key;
   }  printf("Sorted array:\n");
   for(i = 0; i < n; i++)
      printf("%d ", a[i]);
  return 0;}
```

Output:

main.c

```c
#include <stdio.h>

int main()
{
    int a[50], n, i, j, key;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for(i = 1; i < n; i++)
    {
        key = a[i];
        j = i - 1;

        while(j >= 0 && a[j] > key)
        {
            a[j + 1] = a[j];
            j--;
        }

        a[j + 1] = key;
    }

    printf("Sorted array:\n");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

Output

```
Enter number of elements: 5
Enter elements:
41 22 10 3 1
Sorted array:
1 3 10 22 41

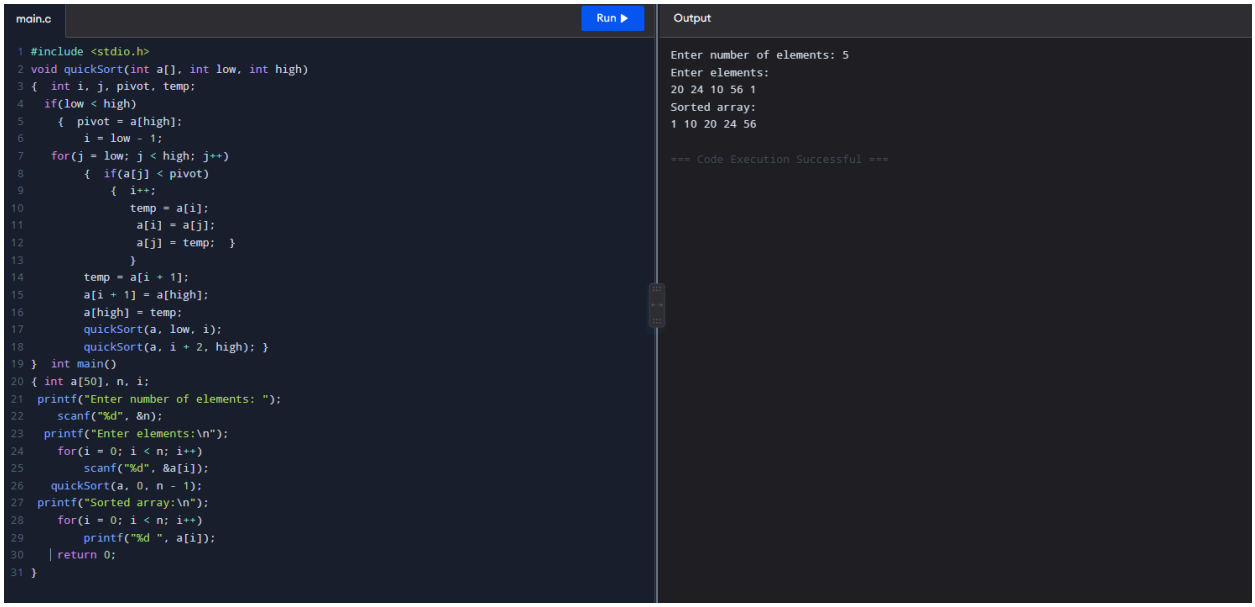
=== Code Execution Successful ===
```

Program: Quick sort

```
 #include <stdio.h>
void quickSort(int a[], int low, int high)
{  int i, j, pivot, temp;
  if(low < high)
    {  pivot = a[high];
       i = low - 1;
   for(j = low; j < high; j++)
       {  if(a[j] < pivot)
          {  i++;
             temp = a[i];
              a[i] = a[j];
              a[j] = temp;  }
             }
       temp = a[i + 1];
       a[i + 1] = a[high];
       a[high] = temp;
       quickSort(a, low, i);
```

```
        quickSort(a, i + 2, high); }
}  int main()
{ int a[50], n, i;
 printf("Enter number of elements: ");
    scanf("%d", &n);
  printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);
   quickSort(a, 0, n - 1);
 printf("Sorted array:\n");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);
    return 0;
}
```

Output:



Program: Merge sort

```
#include <stdio.h>
void merge(int a[], int low, int mid, int high)
{ int temp[50];
    int i = low, j = mid + 1, k = low;
  while(i <= mid && j <= high)
      if(a[i] <= a[j])
            temp[k++] = a[i++];
```

```
        else
            temp[k++] = a[j++]; }
       while(i <= mid)
        temp[k++] = a[i++];
   while(j <= high)
        temp[k++] = a[j++];
    for(i = low; i <= high; i++)
        a[i] = temp[i];  }
void mergeSort(int a[], int low, int high)
{  int mid;
 if(low < high)
    {  mid = (low + high) / 2;
 mergeSort(a, low, mid);
 merge(a, low, mid, high); }
}   int main()
{  int a[50], n, i;
   printf("Enter number of elements: ");
    scanf("%d", &n);
   printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);
 mergeSort(a, 0, n - 1);
 printf("Sorted array:\n");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);
  return 0; }
```

Output:

```c
#include <stdio.h>
void merge(int a[], int low, int mid, int high)
{ int temp[50];
    int i = low, j = mid + 1, k = low;
  while(i <= mid && j <= high)
     if(a[i] <= a[j])
          temp[k++] = a[i++];
       else
          temp[k++] = a[j++]; }
     while(i <= mid)
       temp[k++] = a[i++];
   while(j <= high)
       temp[k++] = a[j++];
    for(i = low; i <= high; i++)
       a[i] = temp[i];  }
void mergeSort(int a[], int low, int high)
{  int mid;
 if(low < high)
    {  mid = (low + high) / 2;
 mergeSort(a, low, mid);
 merge(a, low, mid, high); }
}   int main()
{  int a[50], n, i;
   printf("Enter number of elements: ");
    scanf("%d", &n);
   printf("Enter elements:\n");
    for(i = 0; i < n; i++)
       scanf("%d", &a[i]);
 mergeSort(a, 0, n - 1);
 printf("Sorted array:\n");
    for(i = 0; i < n; i++)
       printf("%d ", a[i]);
  return 0; }
```

```
Enter number of elements: 5
Enter elements:
21 11 54 10 2
Sorted array:
2 10 11 21 54

=== Code Execution Successful ===
```